

Project Design

PocketSmart AI – Your Smart Budget & Recommendation Assistant

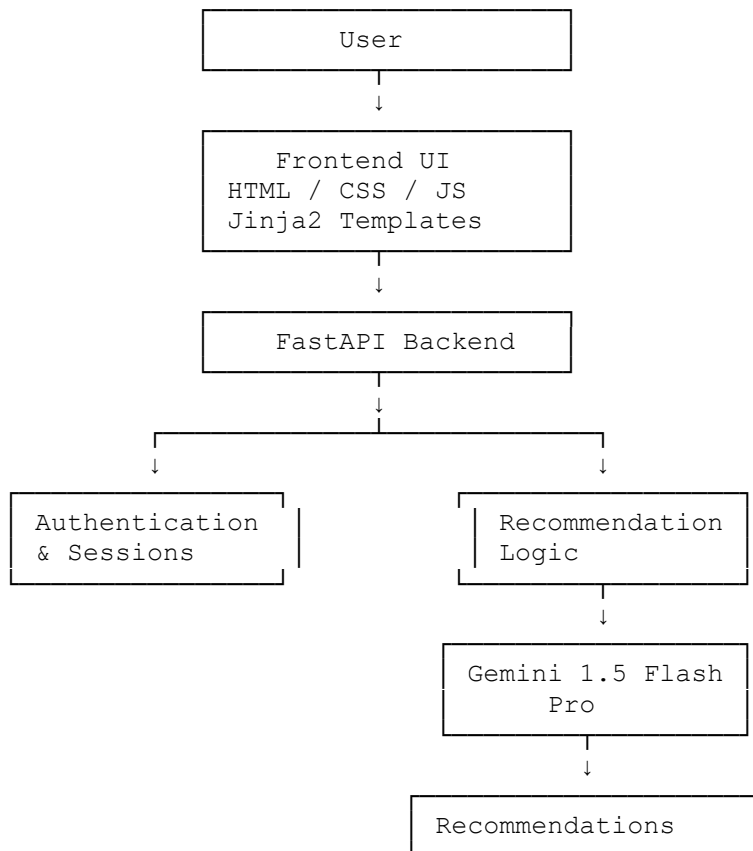
1. System Design

PocketSmart AI follows a modular web application architecture.

The major layers are:

1. User Interface
2. FastAPI Backend
3. Recommendation Service
4. Gemini AI Layer
5. Authentication & Session Management
6. External Product/Service Platforms

2. High-Level Architecture



3. Frontend Design

The frontend provides interfaces for:

- Home page
- Registration
- Login
- User dashboard
- Home Planner
- Party Planner
- Jewelry Planner
- Recommendation pages
- Recommendation history
- Testimonials

The interface is designed to display AI-generated recommendations in structured layouts.

4. Backend Design

FastAPI acts as the main backend framework.

Important routes include:

```
/register
/login
/logout
/token

/generate-home
/generate-party
/generate-jewelry

/session-info
/session-data
/recommendations-details
/history
/startup
```

5. AI Design

Gemini 1.5 Flash Pro processes:

- Text input
- Budget information
- User preferences
- Category information
- Optional images

The AI then generates contextual recommendations.

6. Planner Design

Home Planner

```
Budget
+
Room Details
+
Product Requirements
↓
Gemini AI
↓
Home Recommendations
```

Party Planner

```
Budget
+
Guest Count
+
Event Type
+
Venue Details
↓
Gemini AI
↓
Party Recommendations
```

Jewelry Planner

```
Budget
+
Occasion
+
Style
+
Optional Outfit Image
↓
Gemini AI
↓
Jewelry Recommendations
```

7. Database / Session Design

The system maintains user-related information through authentication and session management.

Session-related endpoints provide information required for personalized interaction and recommendation tracking.

8. External Platform Design

The project is designed to source or simulate product/service information from platforms such as:

- Amazon
- Flipkart
- IKEA
- Swiggy
- Zomato
- OYO

9. Security Design

Security-related components include:

- User authentication
- JWT tokens
- Protected routes
- Session management
- Environment variables for sensitive configuration
- CORS configuration